

Detailed Lab Report (Full Version)

Lab Title:

Network Protocol Attacks Lab (SMB Relay & MITM Analysis)

Objectives:

- Perform SMB Relay attack using Responder
 - Conduct MITM attack using Ettercap
 - Analyze captured traffic using Wireshark
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Tools Used:

- Kali Linux
 - Responder
 - Ettercap
 - Wireshark
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Lab Environment:

- Attacker Machine: Kali Linux
 - Target Machine: TryHackMe VM
 - Network Range: 192.168.1.0/24
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Attack 1: SMB Relay (Responder)

Description:

Responder was used to poison LLMNR/NBT-NS requests and capture NTLM authentication hashes.

Steps Performed:

1. Launched Responder on network interface
2. Waited for victim request (SMB/HTTP)
3. Captured NTLM hash

Result:

Attack ID	Technique	Status	Outcome
015	SMB Relay	Success	NTLM Hash

Findings:

- NTLMv2 hash successfully captured
- Target system vulnerable to LLMNR poisoning
- SMB signing not enforced

Attack 2: MITM (ARP Spoofing using Ettercap)

Description:

Ettercap was used to perform ARP spoofing, positioning the attacker between victim and gateway.

Steps Performed:

1. Enabled IP forwarding
2. Executed ARP spoofing attack
3. Intercepted live traffic

Findings:

- Successful interception of network packets
 - Observed HTTP and DNS traffic
 - Potential credential exposure in plaintext
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Traffic Analysis (Wireshark)

Observations:

- NTLM authentication packets detected
 - DNS spoofing responses identified
 - Unencrypted HTTP traffic visible
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Security Issues Identified:

- LLMNR/NBT-NS enabled
 - Lack of SMB signing
 - Unencrypted communication (HTTP)
 - Susceptibility to ARP spoofing
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Recommendations:

Network Hardening:

- Disable LLMNR and NBT-NS
- Enable SMB signing
- Use HTTPS instead of HTTP

Monitoring:

- Deploy IDS/IPS systems
- Monitor ARP traffic anomalies

Authentication Security:

- Use Kerberos instead of NTLM
 - Implement multi-factor authentication
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Conclusion:

This lab clearly demonstrated how real-world network attacks such as SMB relay and Man-in-the-Middle (MITM) can compromise systems when security controls are weak. Using Responder, NTLM authentication hashes were captured by exploiting LLMNR/NBT-NS

misconfigurations, showing how legacy protocols can leak credentials. Ettercap was then used to perform ARP spoofing, allowing the attacker to intercept and relay traffic between the victim and gateway.

From a practical perspective, this means an attacker on the same network can silently observe or manipulate communication, especially when encryption is not enforced. Analysis with Wireshark confirmed the presence of NTLM authentication exchanges and unencrypted protocols like HTTP, highlighting serious visibility into sensitive data.

Overall, the lab emphasizes that insecure configurations—such as enabled LLMNR, lack of SMB signing, and absence of encrypted communication—create exploitable entry points. Implementing secure protocols, enforcing authentication protections, and continuous network monitoring are essential to defend against these types of attacks.